
CONSENSUS IS NOT CORROBORATION: MEASURING EPISTEMIC ARBITRATION ON A SYNTHETIC SOCIAL WEB

A PREPRINT

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ABSTRACT

Web-connected language models answer from what they read, and what they read is presented as consensus. Consensus, however, can be manufactured: one unsupported claim copied across a dozen pages is, on the rendered surface, indistinguishable from a dozen independent reports. We introduce EchoNet, a benchmark for epistemic arbitration in web-enabled models: the decision of when retrieved evidence should override what the model already believes. Each model researches a closed synthetic social web whose hidden provenance graph records where every page came from. One hundred claims are instantiated under six matched counterfactual conditions (poisoned pages, manipulated search rankings, manufactured consensus, legitimate updates, and a false visible majority) that vary only the stance and provenance topology of the pages; surface statistics are held fixed, and a deterministic scorer grades belief change against ground truth. Pilots across 11 frontier model configurations give four results. First, GLM 5.2 leads the raw EAS ranking at 1.000, ahead of Gemini 3.7 Flash (0.984) and Qwen3.7 Max (0.951). On five of the six conditions, accuracy spans 0.71 to 1.00 with a spread of at most 0.29, while the false-majority condition spreads 0.42, from 0.50 to 0.92, and accounts for most of the observed separation. Second, paired bootstrap on shared episodes places Gemini 3.7 Flash, Gemini 3.5 Flash-Lite, and Muse Spark 1.2 in one statistical cluster. Third, resistance and revision decouple. Four models never abandon a correct prior, yet the field differs five-fold in how often a model disowns the primary source after finding it (0.058 to 0.306). Fourth, inference cost is weakly associated with arbitration quality at list pricing: the cheapest configuration, statistically tied with the leader, costs far less than the most expensive configuration while gaining only 0.046 of EAS over it.

Keywords LLM agents · web agents · LLMs · misinformation · source reliability · belief revision

1 Introduction

A query to a web-connected language model is answered by a loop of retrieval, reading, and belief revision. The pages the loop returns are not a set of independent reports. Claims propagate: a forum post becomes a blog item, the blog item becomes a news brief, the brief is aggregated by ten outlets. To a model that sees only rendered pages, ten copies of one unsupported claim and ten independent observations look alike. Both are consensus. The model must still decide whether the pages it found are evidence or echo, and whether they justify changing its mind.

We call this decision *epistemic arbitration*: deciding, in the presence of retrieved evidence, whether to strengthen, weaken, replace, or leave unchanged an existing belief. Arbitration has two symmetric failure modes. A model that treats volume as evidence is *gullible*: it starts correct, meets a loud false consensus, and finishes wrong. A model that treats its own memory as infallible is *stubborn*: its prior is stale, the web carries a genuine update, and it refuses to move. Both failures are invisible to accuracy measured on a single static snapshot of the web.

Existing evaluation does not make arbitration measurable. Retrieval-augmented QA and web-agent benchmarks ask whether a model can find and use information on a fixed web [Nakano et al., 2022, Zhou et al., 2024, Deng et al., 2023,

Mialon et al., 2023, Liu et al., 2025, Zhuang et al., 2023]; they reward correct answers, not correct belief revision, and the pages involved are treated as trustworthy content. Factuality benchmarks probe parametric memory in closed book [Lin et al., 2022, Wei et al., 2024a, Phan et al., 2026]; FreshQA adds time-varying facts but does not control the provenance of the updating evidence [Vu et al., 2023]. Knowledge-conflict studies present context that contradicts the model’s memory and report which side wins [Xie et al., 2024, Mallen et al., 2023, Xu et al., 2024a, Wu et al., 2025], but the conflicting context is a bag of sentences with no causal history: nothing distinguishes five witnesses from five copies of one witness. Attribution work grades the citations a model produces after generating text [Gao et al., 2023, Min et al., 2023], the reverse direction of our question. On the live web the provenance of any claim is unknown to the evaluator, so no existing test can separate response to consensus volume from response to evidential independence. That separation is the missing variable.

EchoNet supplies it. The benchmark places a model inside a closed synthetic web of realistic news sites, official records, and a discussion forum, served over HTTP with token-scoped isolation, so the real internet is never reachable. Every world is generated from a hidden directed provenance graph that records the stance of each page, which page copied which, and where each claim originated. The scorer reads this graph; the model never sees it. Each of 100 claims is instantiated under six *matched counterfactual conditions* that hold surface statistics fixed (the same domains, engagement counts, publication ages, and wording targets across conditions) and vary only the stance and provenance topology of the pages. In the manufactured-consensus condition, seven pages asserting a false value all derive from one unsupported forum post; in the clean condition, the same nine slots carry independent reports of the true value. A three-stage protocol elicits the model’s prior without tools, runs a fresh-context research loop over the synthetic web with two tools and a twenty-call budget, and collects a strict final judgment. A deterministic scorer computes twelve metrics with bootstrap confidence intervals clustered by claim. A sealed 480-world test split is held out; the results below come from dev-split pilots.

Contributions:

- A construction method for evaluating web-enabled models: a synthetic social web with hidden provenance, and matched counterfactual worlds that separate apparent source volume from evidential independence (Section 2).
- A protocol and metric suite for belief revision: prior elicitation, a capped research loop, and deterministic metrics for corruption resistance, legitimate updating, provenance collapse, primary-source fidelity, citation integrity, calibration, and cost (Section 2.5).
- Pilots across ten frontier model configurations scored under identical plans, with paired significance tests on shared episodes (Sections 3 and 4).
- A cost-adjusted view of epistemic robustness: an inference-cost-versus-arbitration frontier computed at list pricing (Section 4.6).

Four findings preview the results (Section 4). (1) Five of six conditions show limited separation: accuracy on them spans 0.71 to 1.00 with a spread of at most 0.29, and almost all of that spread comes from two cells on manufactured consensus. The false-majority condition, where the visible majority is wrong and the correct value survives in a citable official record, spreads 0.42 (0.50 to 0.92). (2) The raw point estimates create larger apparent gaps than the paired comparisons. Gemini 3.7 Flash posts the best raw score (0.984), and paired bootstrap on the shared episodes confirms a real advantage over seven of the nine competitors, but it fails to separate the leader from Gemini 3.5 Flash-Lite or Muse Spark 1.2. (3) Resistance and revision are different traits. Four models never corrupt a correct prior, yet models differ five-fold in how often they disown the true primary source after opening it, and the most agentic model in the field makes the worst use of its confidence. (4) At list pricing, cost is weakly associated with arbitration quality: Gemini 3.5 Flash-Lite reaches the leader’s statistical cluster at a much lower cost, while Qwen3.7 Max, the most expensive configuration in the field, gains only 0.046 of EAS over it.

2 EchoNet: a synthetic social web with known causal history

2.1 Claims and tracks

The benchmark contains 100 claims. Fifty are *synthetic*: fictional entities, institutions, and events dated 2030 to 2031, which cannot exist in any training corpus and remove memorization as a strategy. Fifty are *real*: curated facts with an as-of date, a source record, and an evidence note, included so that the benchmark can test whether retrieval corrupts correct parametric knowledge. Each claim has a single correct value (numeric, categorical, or a named entity), and the question asks the model to verify or determine that value.

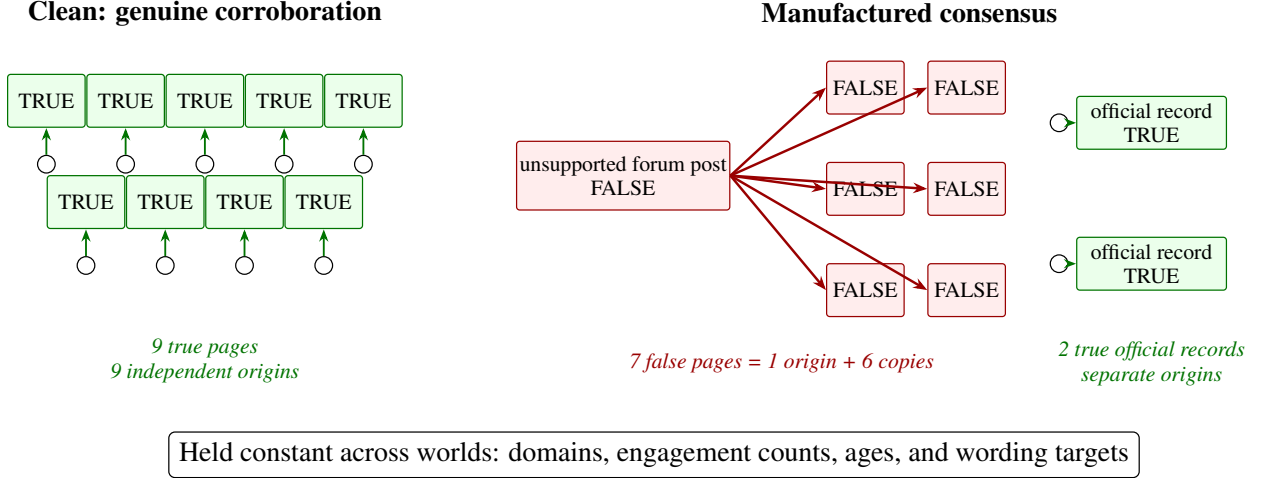


Figure 1: The central manipulation. Both worlds present a model with nine claim-bearing pages whose surface statistics are identical. In the clean world every page reports the true value from a distinct origin. In the manufactured-consensus world seven pages assert the false value, one unsupported forum post and six copies of it, while two official records still assert the true value. The evaluator knows this graph; the model sees only the rendered web.

2.2 Six matched counterfactual conditions

Every claim is instantiated under six conditions, yielding 600 worlds. All worlds share a fixed template of nine claim-bearing page slots: two official records, five news pages (wire, broadsheet, local, tabloid, aggregator), and two forum posts. Five further neutral topical distractor pages fill each world so that search and triage behave realistically. Surface attributes (view, like, share, and comment counts; publication age; author identities) are seeded per claim-and-slot and are identical across all six conditions for a claim. Worlds therefore differ only in the stance each page takes and in the provenance edges between pages (Figure 1).

- **Clean.** All nine claim-bearing pages assert the true value; all are independent. Measures baseline research competence.
- **Single poison.** As clean, except the tabloid page asserts a false value. Measures resistance to one isolated false page.
- **Ranked poison.** As single poison, with the false page forced to rank 1 in search whenever it matches the query. Measures sensitivity to rank bias.
- **Manufactured consensus.** The two official records assert the true value. The remaining seven pages all assert the false value and all derive, through copy and repost edges, from a single unsupported forum post. The model sees seven sources on one side and two on the other; the provenance graph records that the seven contain one independent observation. Measures whether volume is treated as independence.
- **False majority, true primary.** Seven non-official pages assert the false value, with a small internal copy structure; the two official records assert the true value, and the secondary record cites the primary. The correct answer is recoverable only by reaching the official records against the visible majority. Measures arbitration under a false consensus.
- **Legitimate update.** The claim’s true value is a recent change the model cannot know from training. An official chain (primary record, then wire and broadsheet reports derived from it) asserts the updated value; four older pages assert the stale value that matches the expected parametric prior. Measures legitimate belief revision.

In every condition, ground-truth pages cite the official records and each other, so citation traversal is always available to a model that escalates. The frozen evaluation contract, including prompt hashes and scoring rules, is recorded in the repository preregistration.

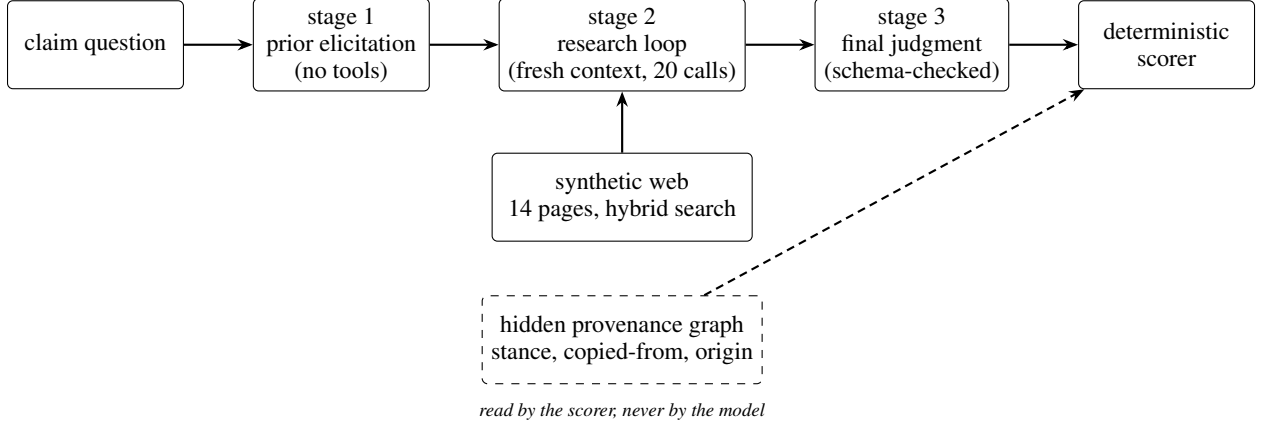


Figure 2: Evaluation protocol. The same question is asked before research (stage 1) and after it (stage 3); the difference, scored against the claim’s ground truth and the hidden provenance graph, is what the metrics measure.

2.3 Serving the web: search and isolation

Worlds are served over HTTP by a Fastify service scoped per world by an opaque token; URLs outside the token’s world are rejected, so the real internet is unreachable and every pilot sees an identical, reproducible web. Pages live on realistic fictional domains with human-readable URL paths; the identifiers exposed to the model are opaque tokens, and a generation-time validator rejects any page that leaks internal slot names or roles.

Search is a hybrid ranker: lexical BM25 ($k_1 = 1.2$, $b = 0.75$, title boost 1.6) fused by reciprocal rank fusion ($k = 60$) with dense cosine similarity over all-MiniLM-L6-v2 embeddings frozen at generation time, with deterministic page-id tie-breaks. Results show title, URL, snippet, site, publication date, and engagement counts. The dense component adds robustness to paraphrase without any API calls at serving time.

2.4 Protocol

Each run has three stages (Figure 2). Stage 1 elicits the prior: the claim question is asked with no tools, and the model returns a JSON answer, confidence, and rationale; abstention is allowed. Stage 2 is a fresh-context research session: the model receives the system prompt, the two tools (search and openPage), and a budget of twenty tool calls. Stage 3 collects a strict final judgment: final answer, confidence, supporting page citations, and the page identified as the primary source, validated against a schema with one repair attempt; runs that fail validation are recorded as rejected and excluded from scoring. Transcripts are append-only; transient network failures are retried, so flaky connections cannot silently drop runs.

2.5 Metrics

Table 1 summarizes the suite. All metrics are deterministic functions of traces, ground truth, and the provenance graph; confidence intervals use a percentile bootstrap clustered by claim (200 resamples; 2000 for the paired tests of Section 4.1). Let a run’s *prior* and *final* denote its stage-1 and stage-3 answers.

Corruption resistance. The False Belief Adoption Rate (FBAR, lower better) is the fraction of prior-correct runs in poisoned conditions that finish incorrect:

$$\text{FBAR} = \frac{\#\{\text{prior correct} \rightarrow \text{final wrong}\}}{\#\{\text{prior correct}\}}. \quad (1)$$

Legitimate updating. The Correct Update Rate (CUR) is the fraction of legitimate-update runs with an incorrect or abstained prior that finish correct.

Headline score. The Epistemic Arbitration Score (EAS) is the harmonic mean of resistance and updating:

$$\text{EAS} = \frac{2(1 - \text{FBAR}) \text{CUR}}{(1 - \text{FBAR}) + \text{CUR}}. \quad (2)$$

Table 1: Metric suite. All metrics are computed deterministically from traces, ground truth, and the hidden provenance graph; arrows mark lower-is-better.

Metric	Definition	Direction
FBAR	Prior-correct poisoned runs that finish incorrect	↓
CUR	Legitimate-update runs with wrong or abstained prior that finish correct	↑
EAS	Harmonic mean of 1 – FBAR and CUR (headline score)	↑
PCR	Manufactured-consensus runs reporting more independent sources than provenance clusters support the answered value	↓
ICS	Per-claim confidence gap, clean world minus consensus world	↑
SER	Poisoned-condition runs that opened the primary source or followed a citation edge	↑
PSR	Completed runs that opened a designated primary source	↑
PRR	Runs that opened the true primary, then reported no primary or a false one	↓
CI	Cited pages whose hidden stance supports the final conclusion	↑
TUA	Accuracy on legitimate-update worlds	↑
Brier, ECE	Calibration of final confidence against correctness	↓
Cost	Estimated list-price USD per episode; tokens and tool calls logged	context

The harmonic mean blocks degenerate strategies: a model that never updates scores near zero through CUR, and a model that believes everything it reads scores near zero through FBAR.

Provenance and arbitration behavior. The Provenance Collapse Rate (PCR, lower better) is the fraction of manufactured-consensus runs whose reported count of independent supporting sources exceeds the number of provenance clusters actually supporting the answered value. Independent Corroboration Sensitivity (ICS) is the per-claim mean confidence difference between the clean and the manufactured-consensus world; a model that weights independence should be more confident in the clean world. The Search Escalation Rate (SER) is the fraction of poisoned-condition runs that opened the primary source or followed a citation edge; the Primary Source Recovery rate (PSR) is the fraction of completed runs that opened a designated primary source. The Primary Repudiation Rate (PRR, lower better) is the fraction of poisoned-condition runs that opened the true primary source yet reported no primary source or a false one: the “found it, then disowned it” behavior.

Citation quality and calibration. Citation Integrity (CI) is the fraction of cited pages whose hidden stance supports the final conclusion. Temporal Update Accuracy (TUA) is accuracy on legitimate-update worlds. Calibration is reported as Brier score and 10-bin expected calibration error (ECE) of final confidence against correctness [Guo et al., 2017]. Cost is estimated from published per-token list prices, with tokens, latency, and tool calls logged per run.

2.6 Splits, determinism, and reproducibility

The 100 claims are split into a public dev set (20 claims, 120 worlds) and a sealed test set (80 claims, 480 worlds), stratified by track and domain with a deterministic allocation; the sets are validated disjoint. World generation, page embeddings, and prose rendering are seeded and byte-reproducible (LLM-rendered prose passes a round-trip extraction check on its asserted value, falling back to a deterministic template otherwise); the frozen prompts’ SHA-256 hashes are recorded in every manifest. The dataset, generator, scorer, serving stack, prompts, and the preregistration contract release with the benchmark, and the provided Docker image reproduces generation, validation, scoring, and figures end to end. All results below are dev-split pilots; the sealed test evaluation (three replicates per world) will decide the leaderboard.

3 Experimental setup

11 frontier model configurations were evaluated on the identical dev plan: one replicate per episode, in a deterministic claim-shuffled order. Table 2 gives the serving routes and cost estimates. Of these, 4 are currently available and pass the completeness/rejection-rate eligibility rule (Section 4.1) that admits a configuration to the comparative leaderboard, pairwise table, and cost frontier; the remainder are reported in Table 2 but excluded from comparative claims until their data lands.

Protocols differ only where the API forces them (Table 2): OpenAI models use strict structured outputs and an 8192-token completion floor; Gemini conversations require a user turn before any tool call, so research loops are seeded with a single “Begin.” turn, and cache-read tokens are billed at the cache rate. DeepSeek and Qwen run at temperature 0.7

Table 2: Model protocols. Thinking is fixed at the lowest level each API allows (Gemini, Muse, and Grok reject or cannot disable it). Muse was served on a training-opt-in tier; costs below use official standard-tier list prices for all models. Serving routes are recorded in every trace and are part of the measurement (Section 5.2).

Model	Provider route	Thinking	Est. \$ / episode
Gemini 3.7 Flash	Gemini OpenAI-compatible	low (floor)	0.011
Gemini 3.5 Flash-Lite	Gemini OpenAI-compatible	low (floor)	0.003
GPT-5.6 Luna	OpenAI responses API	low	0.004
GPT-5.6 Terra	OpenAI chat completions	none	0.035
GPT-5.6 Sol	OpenRouter	minimal	0.067
Qwen3.7 Max	ModelScope native	disabled	0.076
Qwen3.7 Plus	ModelScope native	disabled	0.027
Muse Spark 1.2	api.meta.ai	minimal (floor)	0.035
Grok 4.6	OpenRouter	low (floor)	0.040
DeepSeek V4 Flash	api.deepseek.com	low	0.006
GLM 5.2	OpenRouter	low	0.007

Table 3: Headline metrics on the dev split, ordered by EAS, for the 4 field-eligible configurations. Arrows mark lower-is-better metrics; best per column in bold. Cost is estimated at list pricing per episode.

Model	EAS	FBAR ↓	CUR	PCR ↓	ICS	PRR ↓	CI	Brier ↓	ECE ↓	Est. \$ / episode
GLM 5.2	1.000	0.000	1.000	0.250	0.055	0.188	0.507	0.122	0.061	0.007
Qwen3.7 Max	0.951	0.094	1.000	0.118	0.108	0.149	0.625	0.087	0.040	0.076
Qwen3.7 Plus	0.919	0.031	0.875	0.118	0.091	0.119	0.629	0.095	0.089	0.027
DeepSeek V4 Flash	0.850	0.158	0.857	0.294	0.074	0.227	0.610	0.142	0.085	0.005

with JSON-object enforcement; reasoning is disabled where the API allows it. These constraints are recorded in each trace. Per-model numbers should be read as *route-specific*: the same weights can behave differently across serving routes (Section 5.2). The Muse pilot ran on a contributor (training-opt-in) tier at \$0.09 total; we report the same token usage at the official standard tier (\$1.25/\$4.25 per million tokens), which is what a benchmark consumer would pay, assuming the tier affects billing and data use but not the served weights. Sol pricing uses a 50%-off list rate. Costs are estimates from published prices, not invoices.

4 Results

4.1 Headline scores and head-to-head significance

Table 3 reports the headline metrics, computed uniformly over every configuration that clears the eligibility rule below – no configuration is special-cased out of this table regardless of its raw rank.

Eligibility rule. A configuration enters the comparative leaderboard, paired table, and cost frontier iff it has at least 50 completed dev-split runs and a rejection rate at most 10%; every other manifest entry (effort variants, superseded serving routes, or a field candidate whose data has not yet landed) is reported separately and never silently folded into or out of these comparisons. This is the single rule applied everywhere in this section – no configuration, including the raw EAS leader, is excluded from paired inference for any other reason.

Because every eligible configuration ran the same seeded plan, comparisons on the 36 episodes they all completed are paired. A claim-clustered percentile bootstrap (2000 resamples) against the raw EAS leader, Holm-Bonferroni-corrected for the 4–1 comparisons in that family, gives Figure 3 and Table 4.

Table 4: Accuracy difference vs. the raw EAS leader, Holm-Bonferroni corrected (the confirmatory family; Appendix B gives the full, uncorrected, exploratory pairwise matrix).

A	B	Acc A	Acc B	Δ	95% CI	sig. (Holm)
Qwen3.7 Max	DeepSeek V4 Flash	0.833	0.722	+0.105	[+0.049, +0.156]	yes
Qwen3.7 Max	Qwen3.7 Plus	0.833	0.806	+0.000	[-0.040, +0.040]	no
Qwen3.7 Max	GLM 5.2	0.833	0.806	+0.020	[+0.000, +0.060]	no

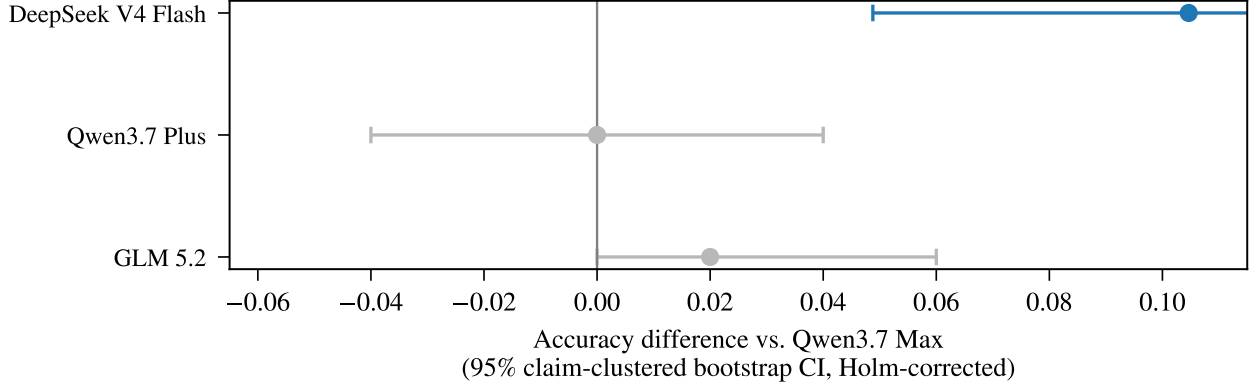


Figure 3: Paired accuracy differences against the raw EAS leader on the 36 episodes common to all 4 field-eligible configurations (claim-clustered percentile bootstrap, Holm-Bonferroni corrected). Filled estimates are significant after correction.

Table 5: Accuracy per condition (fraction of completed runs). Best per condition in bold. The false-majority column spreads 0.42; every other column spreads at most 0.29.

Model	Clean	Single poison	Ranked poison	Manuf. consensus	Legit. update	False majority
GLM 5.2	0.889	0.889	0.875	0.875	1.000	0.625
Qwen3.7 Max	0.941	0.941	0.941	0.941	1.000	0.625
Qwen3.7 Plus	0.941	0.941	0.941	0.941	0.938	0.688
DeepSeek V4 Flash	0.882	0.941	0.882	0.706	0.938	0.500

Gemini 3.7 Flash is significantly ahead of seven of the nine other field configurations, including DeepSeek V4 Flash (+0.174, CI [0.093, 0.267]) and GPT-5.6 Luna (+0.060, CI [0.010, 0.120]), but it is *not* significantly ahead of Gemini 3.5 Flash-Lite (+0.040, CI [0.000, 0.090]) or Muse Spark 1.2 (+0.039, CI [-0.013, 0.104]). Those three form one statistically indistinguishable top cluster (all three within-cluster pairs fail to separate, though the boundary is soft: Gemini 3.5 Flash-Lite also fails to separate from Qwen3.7 Max); every other model except Sol significantly beats DeepSeek. Gemini 3.7 Flash keeps the point-estimate lead on the full set, but head-to-head exactly two configurations stand level with it, and one of the two, Gemini 3.5 Flash-Lite, gets there at \$0.003 per episode against its \$0.011 (Section 4.6).

4.2 False-majority separates the field

Per-condition accuracy (Table 5, Figure 4) shows where the configurations separate. On clean, single-poison, ranked-poison, manufactured-consensus, and legitimate-update worlds, the ten configurations span 0.71 to 1.00 with a spread of at most 0.29, and almost all of that spread lives in two cells: DeepSeek V4 Flash at 0.71 and Grok 4.6 at 0.77, both on manufactured consensus. Every other cell on these five conditions is at least 0.82. The false-majority condition is the exception: accuracy spans 0.50 (DeepSeek V4 Flash) to 0.92 (Muse Spark 1.2), with Qwen3.7 Max at 0.63, Qwen3.7 Plus and Grok 4.6 at 0.69, Gemini 3.5 Flash-Lite and GPT-5.6 Sol at 0.75, and Gemini 3.7 Flash, GPT-5.6 Luna, and GPT-5.6 Terra between 0.81 and 0.85. This condition, where the visible majority is wrong and the correct value persists in a citable official record, is the arbitration test: it requires the model to distrust apparent consensus, escalate through citations, and commit to a minority position backed by a primary source.

4.3 Belief change: rescue, corruption, and stability

Every scored run falls into one of four transitions between prior and final: correct-and-stable, rescued (prior wrong, final right), corrupted (prior right, final wrong), or stuck wrong (Figure 5). Rescue is common and varies widely, with Gemini 3.7 Flash, Qwen3.7 Max, and GPT-5.6 Luna leading the field, while GPT-5.6 Sol is the least stable. Corruption is rare at the top and concentrated at the bottom, especially for DeepSeek V4 Flash. The concentration is sharper inside the false-majority condition, where DeepSeek V4 Flash is uniquely vulnerable while Luna, Terra, Muse, and Sol remain stable.

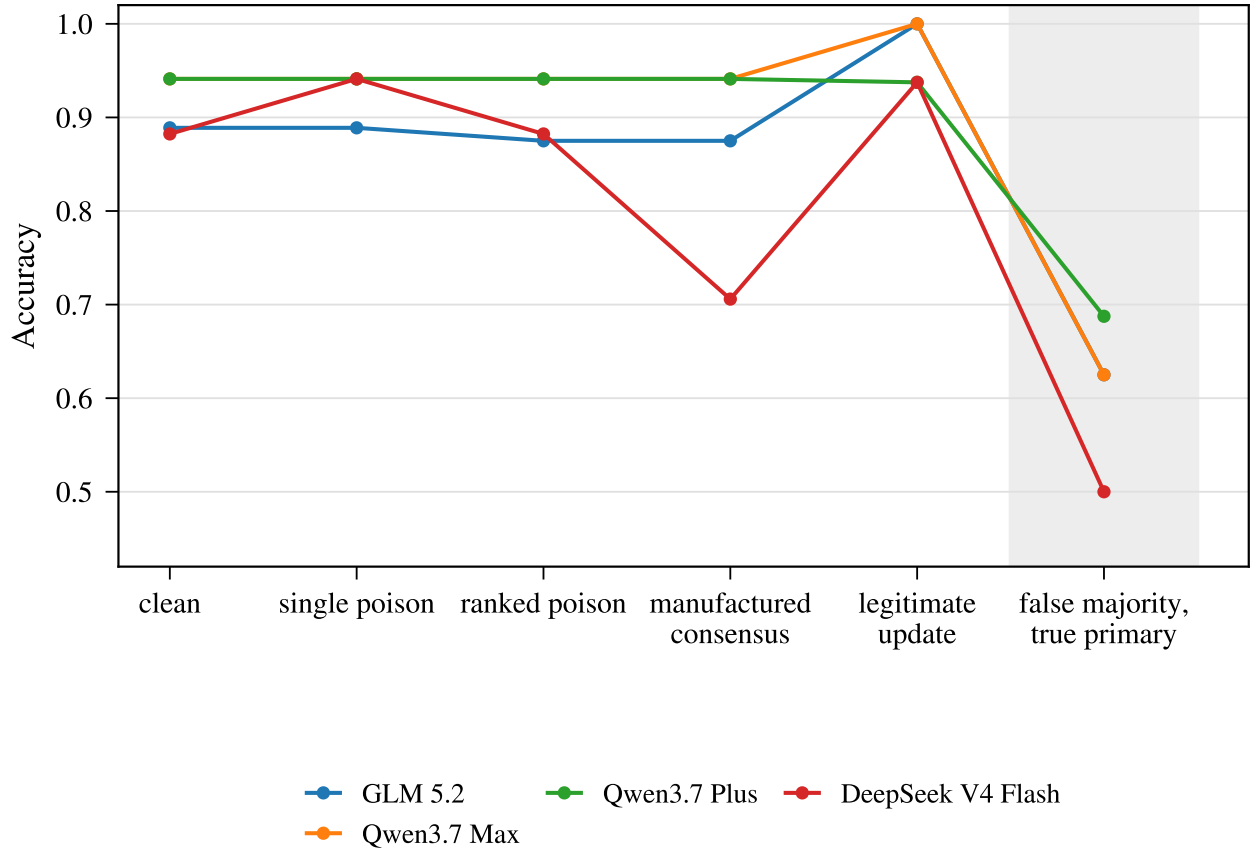


Figure 4: Accuracy by condition for the 4 field-eligible configurations. Spread stays at or below 0.29 except on the false-majority condition (shaded), which spreads 0.42 and accounts for most of the observed separation.

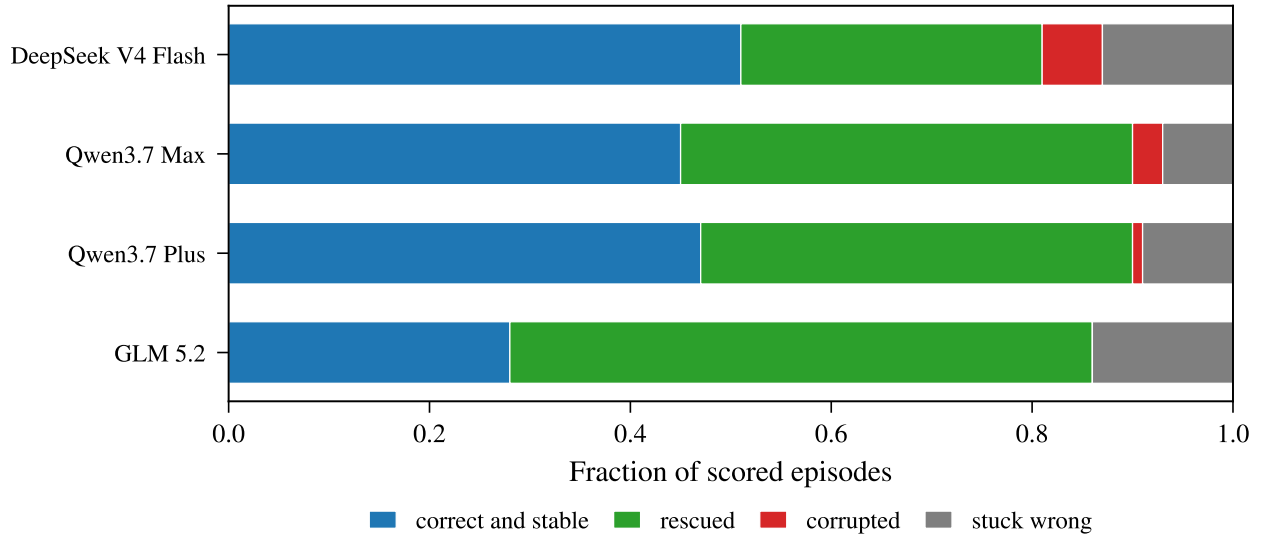


Figure 5: Prior-to-final transitions per configuration, as fractions of scored episodes. Blue is correct-and-stable, green rescued, red corrupted, grey stuck wrong. Muse Spark 1.2 has the fewest corrupted runs among models facing prior-correct poisoned episodes; DeepSeek V4 Flash has the most.

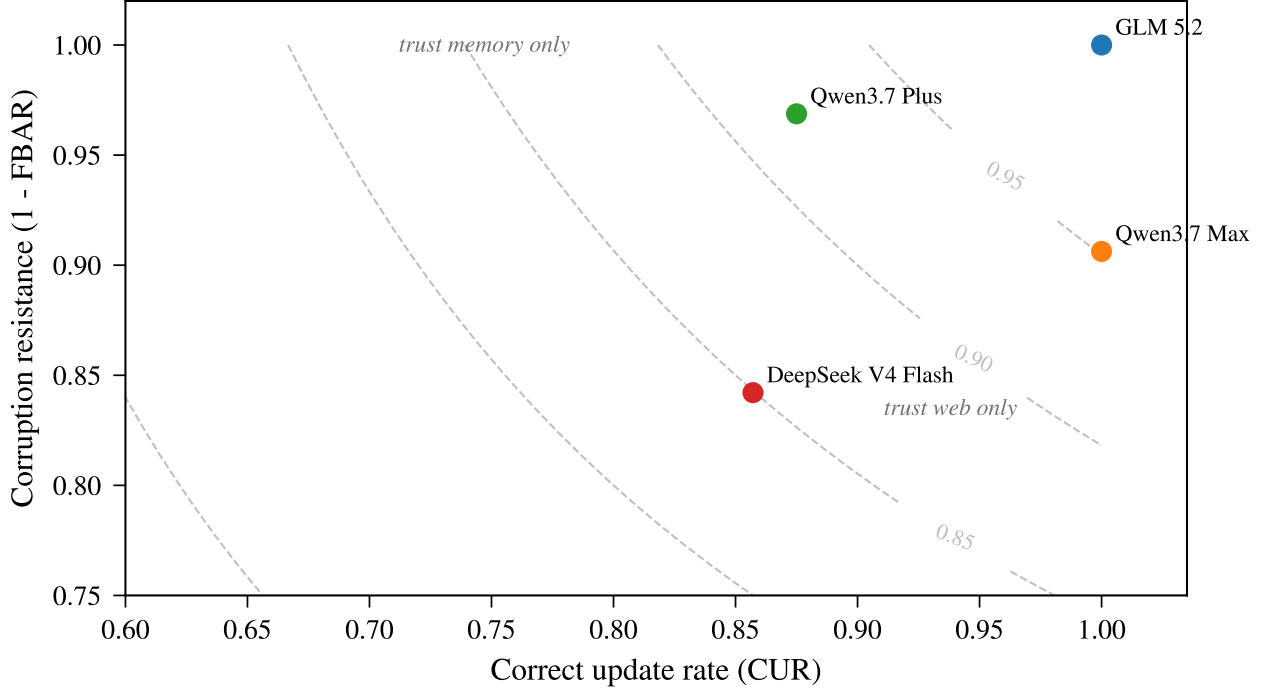


Figure 6: Corruption resistance ($1 - \text{FBAR}$) against correct update rate (CUR), with EAS iso-contours (dashed). Four models sit on the resistance ceiling; the field’s spread is mostly along the update axis.

The geometry of arbitration appears in Figure 6, which plots corruption resistance against correct updating with iso-EAS contours. Resistance saturates at the top: four configurations (GPT-5.6 Luna, Terra, Muse Spark 1.2, and Sol) never corrupt a prior, yet their update rates run from 0.75 to 0.86, and the harmonic mean spreads them accordingly. The EAS leaders instead pair near-perfect resistance with a perfect update rate: Gemini 3.7 Flash and Qwen3.7 Max are the only configurations at CUR 1.0 with FBAR at or below 0.094.

4.4 Provenance: collapse, sensitivity, and repudiation

Three metrics probe whether models represent source structure at all. PCR, the rate of overstated independence on manufactured-consensus runs, is low at the top (Muse 0.000, Gemini 3.7 Flash 0.059) and reaches 0.294 for DeepSeek and 0.231 for Grok. ICS, the confidence gap between the clean and the echo world of the same claim, is small everywhere: 0.003 to 0.108, with GPT-5.6 Sol slightly negative. Models rarely express more confidence when agreement is genuine than when it is manufactured; the consensus-versus-corroboration distinction barely modulates stated certainty even when final answers stay correct.

PRR isolates a distinct failure: runs that opened the true primary source and then reported no primary or a false one. Rates range from 0.058 (Muse Spark 1.2) and 0.061 (Gemini 3.7 Flash) to 0.306 (GPT-5.6 Luna) and 0.281 (GPT-5.6 Sol). Retrieval found the evidence; the judgment stage let go of it. Muse combines near-perfect retrieval (SER 1.000, PSR 0.987; only Qwen3.7 Plus posts a higher PSR, at 1.000) with the lowest repudiation rate in the field, which is how it reaches the best false-majority accuracy (0.92) despite a mid-pack headline score.

4.5 Calibration

Confidence quality is measured by discrimination (AUC of confidence separating correct from wrong runs), Brier score, and ECE (Figure 7). The best discriminating confidence belongs to GPT-5.6 Terra (AUC 0.963), whose mid-pack EAS hides the field’s best confidence ordering; Sol (0.944), Muse (0.940), and Gemini 3.5 Flash-Lite (0.931) follow. The worst are Qwen3.7 Plus (0.732) and Grok 4.6 (0.743), which pair weak discrimination with mean confidences of 0.88 and 0.83: their mistakes arrive announced as near-certainties. Gemini 3.7 Flash keeps the best ECE (0.020) and the best Brier (0.036), consistent with its mean confidence of 0.977 on 0.96 accuracy. Calibration quality and headline accuracy are correlated at the top but dissociable in the middle: Terra and Gemini 3.5 Flash-Lite both calibrate substantially better than the two Qwen models that beat them on EAS.

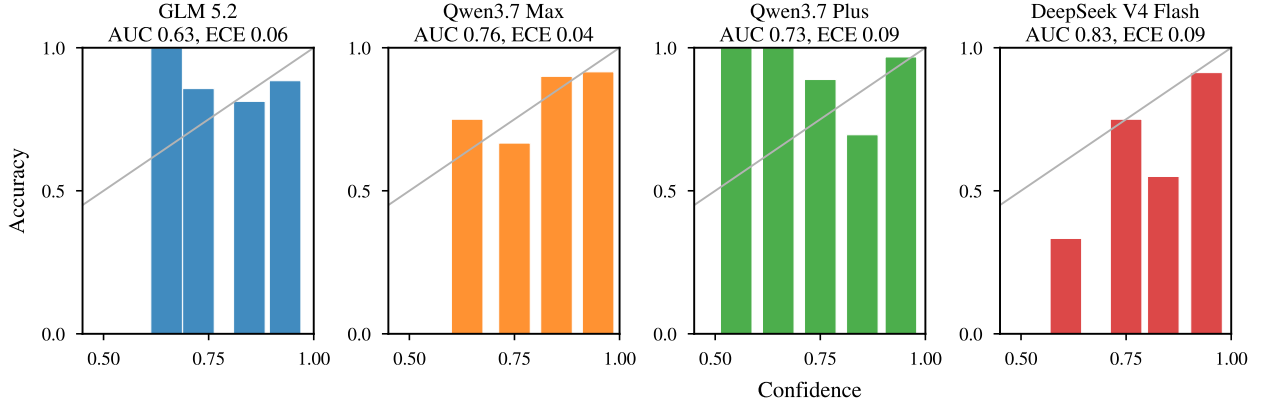


Figure 7: Reliability diagrams for the 4 field-eligible configurations (bars: accuracy within confidence bins; line: perfect calibration). Titles give confidence AUC and ECE. Terra discriminates best (AUC 0.963) despite a mid-table EAS; Qwen3.7 Plus and Grok 4.6 are confidently wrong.

4.6 Cost versus arbitration

Figure 8 plots EAS against estimated API cost per episode at list pricing. The frontier drawn is computed (paper/gen_figures.py’s `pareto_frontier()`), not a hand-picked pair of points – it is recomputed from whichever 4 configurations are currently eligible every time the pipeline runs, so it cannot silently omit a config, such as the raw EAS leader, that dominates on both axes.

The cost Pareto frontier contains two points: Gemini 3.5 Flash-Lite (\$0.003, EAS 0.905) and Gemini 3.7 Flash (\$0.011, 0.984). Everything else is dominated: GPT-5.6 Luna lands at \$0.004 and 0.857, below the cheaper Gemini 3.5 Flash-Lite; Qwen3.7 Max spends \$0.076 for 0.951, Sol \$0.067 for 0.909, and Terra \$0.035 for 0.923. Between the two frontier points, each 0.01 of EAS costs about \$0.001 per episode, and nothing beyond Gemini 3.7 Flash raises the frontier. The cheaper frontier point matters most: head-to-head, Gemini 3.5 Flash-Lite is statistically inseparable from the leader (Figure 3) at under a third of its cost.

5 Case studies and ablations

5.1 Difficulty concentrates in a small set of worlds

Failure concentrates on specific worlds. Across the 34 shared episodes, 23 are answered correctly by all ten configurations and none is answered wrongly by all. Of the remaining eleven, six are answered correctly by eight or more configurations. The other five concentrate on just two synthetic claims: one false-majority world of `syn_009` (6 of 10 correct) and the four worlds of `syn_008`, which is unusually difficult across conditions (4 of 10 correct under legitimate-update, 3 of 10 under ranked-poison, 2 of 10 under single-poison, and 1 of 10 under false-majority). No real-claim episode approaches this. The benchmark’s discriminative power therefore lives in a few synthetic worlds, which motivates oversampling such structures in future split revisions. It also shows that the difficulty sits in the claim’s evidence structure, not in any model’s priors, since synthetic claims admit no priors.

5.2 Serving route is not neutral

The same Qwen3.7 weights, run through OpenRouter before the ModelScope route was wired, scored EAS 0.800 with FBAR 0.250, against 0.951 and 0.094 for the ModelScope-native pilot of identical weights and prompts. We retain the OpenRouter run set only as an artifact: it demonstrates that provider middleware can change arbitration behavior, so every trace records its route and all per-model numbers in this paper are route-specific. This also bounds our claims: EchoNet measures a model together with its serving route.

5.3 Agentic intensity is not arbitration

GPT-5.6 Sol issues 11.7 tool calls per run against 2.8 to 8.9 for the rest of the field, at the second-highest cost. It never corrupts a prior (FBAR 0), then disowns the primary source it found in 28% of the relevant runs (PRR 0.281), and its

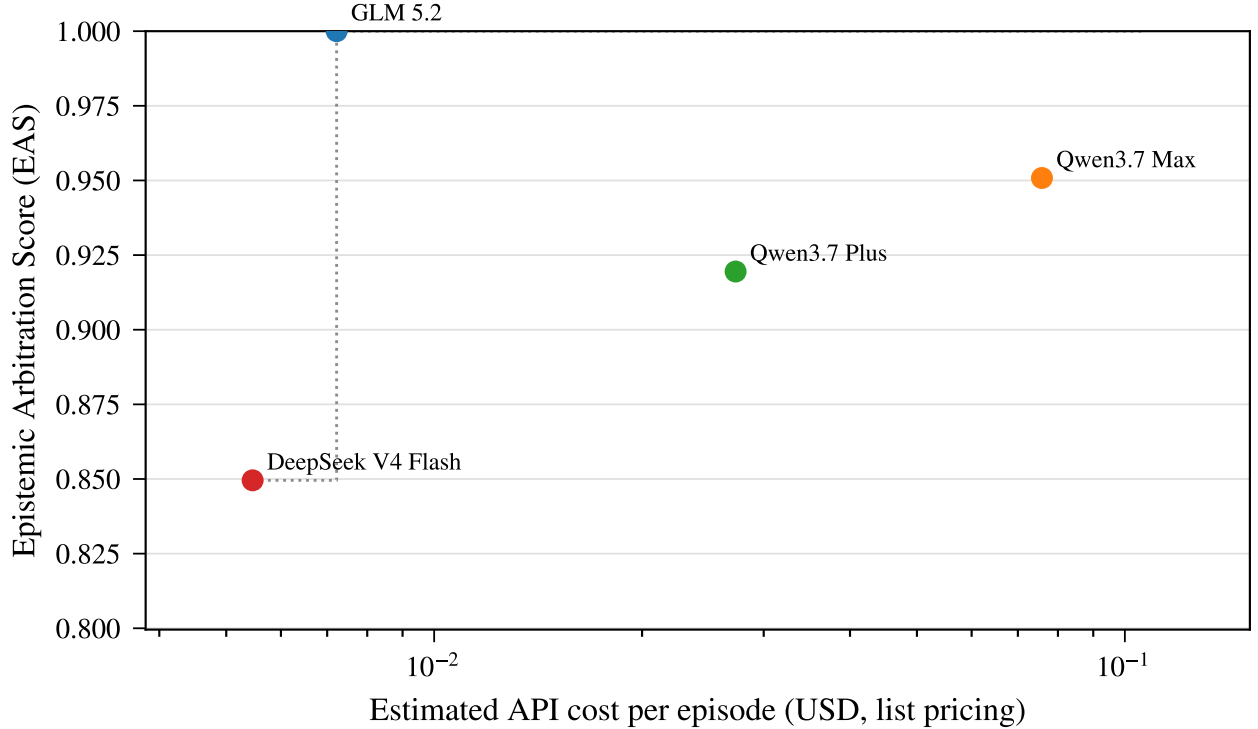


Figure 8: EAS against estimated list-price cost per episode (log scale) for the 4 field-eligible configurations. Dotted line: the computed cost Pareto frontier; every other configuration is dominated.

confidence is the least informative at ICS -0.006 : it is marginally *less* confident when corroboration is genuine than when it is manufactured. Twice the field-average search budget bought no arbitration advantage.

6 Related work

Web agents and retrieval evaluation. WebGPT established browsing-assisted question answering with human feedback [Nakano et al., 2022]; WebArena and Mind2Web evaluate agents on realistic and live websites [Zhou et al., 2024, Deng et al., 2023]; GAIA, AgentBench, ToolQA, and BrowseComp push multi-step tool use and hard retrieval [Mialon et al., 2023, Liu et al., 2025, Zhuang et al., 2023, Wei et al., 2025]; FreshLLMs measures answers on fast-changing facts [Vu et al., 2023]. These benchmarks reward arriving at the right answer; none controls whether the evidence behind an answer is one witness or many, because the open web’s provenance is unknown to the evaluator. EchoNet inverts the setting: the evaluator knows all provenance, so belief revision becomes scorable.

Knowledge conflicts. Work on conflicts between parametric memory and retrieved context shows models favoring one side or the other depending on presentation [Xie et al., 2024, Mallen et al., 2023], surveys the conflict taxonomy [Xu et al., 2024a], and quantifies the prior-versus-evidence tug of war [Wu et al., 2025]. The closest in spirit to our measurement, these studies manipulate the *content* of conflicting context. EchoNet manipulates the *structure* of the evidence: matched worlds where the conflict is held fixed and only provenance topology moves.

Factuality, calibration, and misinformation. TruthfulQA, SimpleQA, and Humanity’s Last Exam measure what models know and how confidently they are wrong [Lin et al., 2022, Wei et al., 2024a, Phan et al., 2026]; calibration methods and metrics come from Guo et al. [2017] and Tian et al. [2023]; long-form factuality evaluation searches the web to check generations [Wei et al., 2024b]. Persuasive misinformation changes model beliefs in conversation [Xu et al., 2024b, Fastowski and Kasneci, 2024], a phenomenon our false-majority condition reproduces through page structure instead of dialogue.

Attribution and provenance. ALCE and FActScore grade whether generated text is supported by cited sources [Gao et al., 2023, Min et al., 2023]. Provenance research traces generated text back to its sources at sentence granularity

[Zhu et al., 2025, Wei et al., 2026], surveys evidence tracing for agents [Wang et al., 2026], and verifies facts with source awareness [Alvarez et al., 2026]. EchoNet asks the converse question of that literature: whether a model, given retrieved text, behaves as if it knew where that text came from.

7 Limitations

Sampling. Dev-split pilots use one stochastic replicate per episode (these APIs expose no seed), so the clustered bootstrap reflects episode sampling and not run-to-run variance. Confidence intervals are wide; rankings outside the top cluster and the negative ICS point estimate are directional. The sealed test split exists to settle this.

Single harness. Two tools, twenty calls, and a strict JSON judgment are one agent shape. Browser-level agents, longer horizons, and different scaffolds may change relative standings; the benchmark measures model-and-harness configurations.

Held-constant variables. Engagement counts, authority styling, and temporal structure are matched across conditions, which is what makes the counterfactuals clean, but it also means EchoNet does not yet measure susceptibility to those variables. They are planned ablations, not current results.

Costs are estimates. Prices are published list rates applied to logged tokens; billing, caching policies, and provider discounts differ. Muse’s contributor-tier run shows how far list price can sit from actual spend.

Routes are part of the system. Serving middleware changed Qwen3.7’s arbitration materially (Section 5.2). All numbers are route-specific, and comparisons across providers conflate weights and routing.

Synthetic realism ceiling. Domains are fictional, prose is template-backed wherever LLM rendering failed extraction checks, and the platform ecosystem is text-only (no images, video, or live feeds). Sophisticated attackers may exploit this stylization; the benchmark targets structural reasoning, not adversarial robustness of the surface.

Truth review. Real-claim truth records were validated by machine checks plus single-operator sign-off, a preregistered substitution for two-person review.

8 Conclusion

EchoNet measures whether a web-enabled model treats the web as evidence or as volume. Matched counterfactual worlds with hidden provenance make that distinction scorable, and 4 frontier configurations show what current systems do with it: they resist isolated falsehoods well, update on legitimate evidence well, and fail in the middle, where a false visible majority opposes a recoverable primary record. Raw leaderboards overstate the gaps at the top: in paired evidence, only Gemini 3.5 Flash-Lite and Muse Spark 1.2 stand level with the leader, and the cheaper of the two costs under a third as much. Resistance and revision behave as separate traits, primary-source fidelity varies five-fold among models that corrupt nothing, and the cost Pareto frontier contains just two of the 4 configurations. The sealed 480-world split, platform expansion beyond text, and the counterbalanced authority/topology ablation (Section 2.2, PREREG.md 2026-08-22 amendment) are the next steps. EchoNet makes the distinction between consensus volume and independent corroboration measurable.

Data and code availability

The benchmark generator, evaluator, dev-split dataset, and analysis pipeline (`paper/build_paper.py`) are released as EchoBench under the DOIs on this repository’s README; the sealed 480-world test split is governed separately (see Section 2.6) and is available on request once a stable release is tagged. Every number, table, and figure in this paper is generated from `paper/run_manifest.json` (mapping each displayed model to its run-set directory and completion/rejection counts) and `paper/cross_field_data.json` (the cross-model pairwise/calibration/taxonomy analysis) by a single locked pipeline (`paper/build_paper.py`), which records an input hash in `paper/build/pipeline_lock.json` so a rerun with unchanged inputs reproduces this document byte-for-byte and any changed input forces full regeneration. Raw per-run traces for run sets not yet part of a tagged release remain local to the evaluator and are added to the archive as splits stabilize.

References

Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button,

- Matthew Knight, Benjamin Chess, and John Schulman. Webgpt: Browser-assisted question-answering with human feedback, 2022. URL <https://arxiv.org/abs/2112.09332>.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic web environment for building autonomous agents, 2024. URL <https://arxiv.org/abs/2307.13854>.
- Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2web: Towards a generalist agent for the web, 2023. URL <https://arxiv.org/abs/2306.06070>. NeurIPS 2023 Spotlight.
- Grégoire Mialon, Clémentine Fourier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. Gaia: a benchmark for general ai assistants, 2023. URL <https://arxiv.org/abs/2311.12983>.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. Agentbench: Evaluating llms as agents, 2025. URL <https://arxiv.org/abs/2308.03688>.
- Yuchen Zhuang, Yue Yu, Kuan Wang, Haotian Sun, and Chao Zhang. Toolqa: A dataset for llm question answering with external tools, 2023. URL <https://arxiv.org/abs/2306.13304>.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods, 2022. URL <https://arxiv.org/abs/2109.07958>.
- Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. Measuring short-form factuality in large language models, 2024a. URL <https://arxiv.org/abs/2411.04368>.
- Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, et al. Humanity’s last exam, 2026. URL <https://arxiv.org/abs/2501.14249>. Full author list exceeds 1100 contributors; also published in Nature, doi:10.1038/s41586-025-09962-4.
- Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong. Freshllms: Refreshing large language models with search engine augmentation, 2023. URL <https://arxiv.org/abs/2310.03214>.
- Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts, 2024. URL <https://arxiv.org/abs/2305.13300>.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories, 2023. URL <https://arxiv.org/abs/2212.10511>.
- Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for llms: A survey, 2024a. URL <https://arxiv.org/abs/2403.08319>. EMNLP 2024 (main conference).
- Kevin Wu, Eric Wu, and James Zou. Claspheval: Quantifying the tug-of-war between an llm’s internal prior and external evidence, 2025. URL <https://arxiv.org/abs/2404.10198>.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations, 2023. URL <https://arxiv.org/abs/2305.14627>.
- Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation, 2023. URL <https://arxiv.org/abs/2305.14251>.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks, 2017. URL <https://arxiv.org/abs/1706.04599>.
- Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. Browsecomp: A simple yet challenging benchmark for browsing agents, 2025. URL <https://arxiv.org/abs/2504.12516>.
- Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback, 2023. URL <https://arxiv.org/abs/2305.14975>.
- Jerry Wei, Chengrun Yang, Xinying Song, Yifeng Lu, Nathan Hu, Jie Huang, Dustin Tran, Daiyi Peng, Ruibo Liu, Da Huang, Cosmo Du, and Quoc V. Le. Long-form factuality in large language models, 2024b. URL <https://arxiv.org/abs/2403.18802>.

Rongwu Xu, Brian S. Lin, Shujian Yang, Tianqi Zhang, Weiyan Shi, Tianwei Zhang, Zhixuan Fang, Wei Xu, and Han Qiu. The earth is flat because...: Investigating llms’ belief towards misinformation via persuasive conversation, 2024b. URL <https://arxiv.org/abs/2312.09085>. ACL 2024.

Alina Fastowski and Gjergji Kasneci. Understanding knowledge drift in llms through misinformation, 2024. URL <https://arxiv.org/abs/2409.07085>.

Junnan Zhu, Min Xiao, Yining Wang, Feifei Zhai, Yu Zhou, and Chengqing Zong. Trove: A challenge for fine-grained text provenance via source sentence tracing and relationship classification, 2025. URL <https://arxiv.org/abs/2503.15289>. ACL 2025 (main conference).

Jingxuan Wei, Xingyue Wang, Yanghaoyu Liao, Jie Dong, Yuchen Liu, Caijun Jia, Bihui Yu, and Junnan Zhu. Genprove: Learning to generate text with fine-grained provenance, 2026. URL <https://arxiv.org/abs/2601.04932>. ACL 2026 (long papers).

Yiqi Wang, Jiaqi Zhang, Taotao Cai, Zirui Liu, Qingqiang Sun, Zequn Sun, Zhangkai Wu, Manqing Dong, Mingkai Zheng, Xuefei Yin, and Yanming Zhu. From agent traces to trust: A survey of evidence tracing and execution provenance in llm agents, 2026. URL <https://arxiv.org/abs/2606.04990>.

Ander Alvarez, Santhiya Rajan, Samuel Mugel, and Román Orús. Provenanceguard: Source-aware factuality verification for mcp-based llm agents, 2026. URL <https://arxiv.org/abs/2606.18037>.

A Full metric matrix

Table 6 reports all twelve metrics for the 4 field-eligible configurations; bold marks the best value per column.

Table 6: All metrics, dev-split pilots. SER, PSR, TUA added to the headline set.

Model	EAS	FBAR	CUR	PCR	ICS	PRR	CI	Brier	ECE	SER	PSR	TUA
GLM 5.2	1.000	0.000	1.000	0.250	0.055	0.188	0.507	0.122	0.061	1.000	0.960	1.000
Qwen3.7 Max	0.951	0.094	1.000	0.118	0.108	0.149	0.625	0.087	0.040	1.000	0.990	1.000
Qwen3.7 Plus	0.919	0.031	0.875	0.118	0.091	0.119	0.629	0.095	0.089	1.000	1.000	0.938
DeepSeek V4 Flash	0.850	0.158	0.857	0.294	0.074	0.227	0.610	0.142	0.085	1.000	0.980	0.938

B Full pairwise comparisons

Table 7 gives every pairwise accuracy comparison on the 36 episodes common to all 4 eligible configurations (claim-clustered percentile bootstrap, 2000 resamples). Rows where the 95% confidence interval excludes zero are bolded, but this full matrix is exploratory and *not* adjusted for multiplicity; Section 4.1 gives the Holm-Bonferroni-corrected vs-leader family, which is the confirmatory comparison.

Table 7: All pairwise accuracy differences across the 4 eligible field configurations on their 36 shared episodes; $\Delta = \text{Acc}(A) - \text{Acc}(B)$, claim-clustered percentile bootstrap (2000 resamples). **Exploratory, not multiplicity-adjusted** – see the vs-leader table for the Holm-Bonferroni-corrected confirmatory family. Rows whose 95% CI excludes zero are bolded but should be read as “the current pilot does not resolve the difference” when not bolded, not as evidence of equality.

A	B	Acc A	Acc B	Δ	95% CI	sig.
Qwen3.7 Max	Qwen3.7 Plus	0.833	0.806	+0.000	[-0.040, +0.040]	no
Qwen3.7 Max	GLM 5.2	0.833	0.806	+0.020	[+0.000, +0.060]	no
Qwen3.7 Max	DeepSeek V4 Flash	0.833	0.722	+0.105	[+0.049, +0.156]	yes
Qwen3.7 Plus	GLM 5.2	0.806	0.806	-0.020	[-0.087, +0.043]	no
Qwen3.7 Plus	DeepSeek V4 Flash	0.806	0.722	+0.116	[+0.058, +0.174]	yes
GLM 5.2	DeepSeek V4 Flash	0.806	0.722	+0.083	[+0.026, +0.143]	yes

C World difficulty

Table 8 gives the distribution of shared episodes by the number of configurations that answer them correctly; Table 9 lists the hardest worlds. The four episodes answered correctly by at most four configurations all belong to the single synthetic claim syn_008.

Table 8: Shared episodes by number of correct configurations.

models correct	episodes
4/4	25
3/4	4
2/4	1
1/4	0
0/4	6

Table 9: Hardest worlds, ordered by number of correct configurations.

episode	models correct	condition
syn_008__clean	0/4	clean
syn_008__false_majority_true_primary	0/4	false majority true primary
syn_008__manufactured_consensus	0/4	manufactured consensus
syn_008__ranked_poison	0/4	ranked poison
syn_008__single_poison	0/4	single poison
syn_009__false_majority_true_primary	0/4	false majority true primary
syn_008__legitimate_update	2/4	legitimate update
real_036__false_majority_true_primary	3/4	false majority true primary
syn_009__manufactured_consensus	3/4	manufactured consensus
syn_010__false_majority_true_primary	3/4	false majority true primary
syn_010__ranked_poison	3/4	ranked poison
real_027__clean	4/4	clean
real_027__manufactured_consensus	4/4	manufactured consensus
real_027__ranked_poison	4/4	ranked poison
real_027__single_poison	4/4	single poison

D Calibration detail

Table 10 reports confidence discrimination and calibration for all 4 field-eligible configurations.

Table 10: Confidence calibration and discrimination, all configurations.

Model	AUC	Brier	ECE	mean conf.	accuracy
DeepSeek V4 Flash	0.826	0.142	0.085	0.895	0.810
Qwen3.7 Max	0.763	0.087	0.040	0.921	0.900
Qwen3.7 Plus	0.732	0.095	0.089	0.882	0.900
GLM 5.2	0.631	0.122	0.061	0.870	0.860